

AI Patient Medical History Summary & Recommendation System

Internship Project Report

Submitted By:

Vishal Vishwas Desai

1. Project Overview

The **AI Patient Medical History Summary & Recommendation System** is a healthcare application developed to help doctors quickly review a patient's medical history before a consultation. In many hospitals, doctors spend valuable time searching through patient records to understand previous illnesses, surgeries, allergies, and medications. This application reduces that effort by automatically retrieving the patient's medical information from a database and presenting it in a clear and easy-to-read format.

The system also generates basic medical recommendations based on the patient's health history, making the consultation process faster and more organized.

2. Problem Statement

During patient checkups, doctors often need to review previous medical records before making decisions. Going through multiple records manually can be time-consuming and may increase the chances of missing important information.

The goal of this project is to create an application that automatically:

- Retrieves patient medical history.
 - Generates a short medical summary.
 - Displays important medical information.
 - Provides recommendations based on the patient's health history.
-

3. Objectives

The main objectives of this project are:

- Develop a system to retrieve patient medical records from a database.
 - Generate a clear and concise summary of the patient's medical history.
 - Provide useful recommendations based on diseases and allergies.
 - Reduce the time doctors spend reviewing patient records.
 - Build a simple and user-friendly interface using Streamlit.
-

4. Technologies Used

Technology Purpose

Python	Backend development
--------	---------------------

Technology Purpose

SQLite Database management

Streamlit Web application interface

ReportLab PDF report generation

VS Code Development environment

5. System Workflow

The application follows these steps:

1. The doctor enters a Patient ID.
 2. The system searches for the patient in the SQLite database.
 3. Patient details are retrieved.
 4. A medical summary is generated automatically.
 5. Recommendations are created based on diseases and allergies.
 6. The doctor can view the complete report on the dashboard.
 7. The report can also be downloaded as a PDF.
-

6. Database Design

The application stores patient information in a SQLite database.

Each patient record contains:

- Patient ID
- Name
- Age
- Gender
- Blood Group
- Allergies
- Previous Surgeries
- Medical Conditions
- Current Medications
- Last Visit Date

- Doctor Notes
-

7. Project Modules

Module 1: Patient Data Retrieval

This module connects to the SQLite database and retrieves patient information using the Patient ID entered by the doctor.

Input

Patient ID

Output

Complete patient medical record.

Module 2: Medical Summary Generation

After retrieving the patient's information, the application generates a structured summary containing important medical details such as diseases, allergies, surgeries, medications, and previous doctor notes.

This helps doctors quickly understand the patient's medical background.

Module 3: Recommendation System

The recommendation module analyzes the patient's diseases and allergies and provides suitable medical suggestions.

For example:

- Monitor blood pressure for hypertension.
 - Check blood sugar regularly for diabetes.
 - Avoid Penicillin if the patient is allergic.
 - Continue prescribed medications.
-

Module 4: PDF Report Generation

The application allows doctors to download the patient's report in PDF format.

The report includes:

- Patient Details
- Medical Summary

- Recommendations

This feature helps maintain patient records and makes it easier to share reports when needed.

8. User Interface

The application is developed using Streamlit.

The dashboard provides:

- Patient ID input
- Patient information
- Medical summary
- Recommendations
- PDF download button

The interface is designed to be simple, clean, and easy to use.

9. Sample Output

Patient Details

- Name: John Doe
- Age: 45
- Blood Group: O+
- Diseases: Hypertension, Migraine
- Allergies: Penicillin, Nuts

Medical Summary

John Doe is a 45-year-old male with a history of hypertension and migraine. He previously underwent an appendectomy in 2015 and is currently taking Aspirin. The patient is allergic to Penicillin and nuts. During the last visit, the doctor advised regular blood pressure monitoring.

Recommendations

- Monitor blood pressure regularly.
- Avoid Penicillin.
- Avoid nuts.
- Continue prescribed medication.
- Schedule regular follow-up visits.

10. Challenges Faced

During development, a few challenges were encountered:

- Designing the database structure for patient records.
- Connecting the Streamlit application with the SQLite database.
- Creating meaningful recommendations using patient history.
- Generating downloadable PDF reports.
- Organizing the project into separate modules for better readability.

These challenges were solved through testing and modular code development.

11. Testing

The application was tested using multiple patient records.

The following scenarios were verified:

- Valid Patient ID retrieval.
- Invalid Patient ID handling.
- Medical summary generation.
- Recommendation generation.
- PDF report download.

All features worked successfully.

12. Future Enhancements

This project can be further improved by adding:

- User authentication for doctors.
 - Electronic Health Record (EHR) integration.
 - AI-powered medical summary generation using NLP.
 - Disease risk prediction using TensorFlow.
 - Appointment scheduling.
 - Cloud database support.
 - Real-time patient monitoring.
-

13. Conclusion

The **AI Patient Medical History Summary & Recommendation System** successfully demonstrates how technology can simplify the process of reviewing patient records during medical consultations. The application retrieves patient information, generates a structured summary, provides useful recommendations, and allows reports to be downloaded in PDF format.

The project improves efficiency, reduces the time required to review patient history, and provides doctors with organized information before patient consultations. It also serves as a strong foundation for future healthcare applications that can integrate artificial intelligence and predictive analytics.